

micro-scale and economies at the macro-scale

- Protecting and sustaining the natural environment

As an example, Google came to the rescue during the Uttarakhand floods of 2013 by developing apps to help rescue teams. It launched a crisis map of the flood-hit state to provide quick information on the places affected, relief centres, medical centres and road closures. It also activated a Person Finder tool—a Web based application that helped feed in the names of the people one was looking for, as well as share information about somebody who had been found or rescued. When there was a match for a person being searched for, it was displayed on this app.

For decades, governments and NGOs have relied on private sector solutions to gather and interpret emergency data for disaster response. But, nowadays, they are depending upon cheaper open source technology.

Proprietary software does everything, from providing images and location based on the Geographic Information System (GIS) to centralising government generated data on a command centre dashboard during a crisis. Such solutions have been around for decades but are expensive.

FOSS: Helping disaster victims

In recent years, especially after the 2010 Haiti earthquake, many cities and NGOs are turning to open source technology, which generally doesn't involve a licence fee to download the software. FOSS disaster management systems are playing an increasingly important role in the rescue and rehabilitation sector as they offer cost-effective coordination for agencies to respond to large scale disasters.

In cases of man-made crises or natural disasters, a range of organisations, websites, open source tools, mobile applications, etc, is available to the public. Some of the FOSS software used in the US during disasters are listed below.

- *CDC widgets and gadgets:* The

Center for Disease Control and Prevention (CDC) now provides a wide range of free software widgets and gadgets focused on public health and safety, for use on computer systems or websites.

- *Frontline SMS:* This is an award winning, free and open source software product that can turn a laptop or mobile phone into a central communications hub. It has proved to be very useful in emergencies.
 - *InSTEDD:* This is focused on the design and use of open source technology tools to help partners enhance collaboration and improve information flow to deliver better critical services to affected populations during disasters.
 - *Integrated Public Alert and Warning System (IPAWS):* This open platform is an initiative of the Federal Emergency Management Agency (FEMA) of the US Department of Homeland Security.
 - *Open ISES:* This provides tools, instruction materials, etc, for emergency services, the community and others.
 - *Real-time Outbreak and Disease Surveillance (RODS):* This is for public health bio-surveillance and is used to collect and analyse data on diseases in real-time. It has been founded by the Climate Prediction Center (CPC) and the National Library of Medicine.
 - *Health Alert Network (HAN):* The Center for Disease Control (CDC) and the Health Alert Network (HAN) provide health alerts, health advisories, updates and information service messages to state and local health officers, public information officers, epidemiologists, HAN coordinators and health care provider organisations across the US.
- Wireless Sensor Networks (WSNs) are widely deployed in disaster management. Managing heterogeneous embedded devices like different processors and memories, as well as operating systems such as embedded Linux, iOS, Android,

along with heterogeneous protocols is very important, to provide efficient solutions for the various challenges in disaster management.

HFOSS plays a key role in disaster management

Humanitarian Free Open Source Software (HFOSS) represents the application of FOSS to fix the problems faced by victims in disaster situations.

FOSS helps to save lives and victims by responding to the following:

- Rapid-onset natural disasters such as tsunamis, earthquakes and pandemics
- Slow-onset natural disasters such as global warming, droughts and famine
- Man-made disasters such as civil wars

Some popular HFOSS solutions used in disaster management are:

- Sahana
- Ushahidi
- Open Street Map
- InSTEDD
- Frontline SMS
- Open MRS
- Rapid SMS
- Google

There are many more institutions contributing to FOSS. These communities work together effectively to provide better software. Most of these products are based on open source platforms and are built from open source software such as Apache, Linux, PHP, Python, MySQL, etc.

The advantages of HFOSS disaster management solutions are:

- No discrimination with respect to access
- Ability to cross technology borders.
- Can easily empower the local people
- Lower cost
- Transparency and neutrality
- Rapid localisation and adaptability
- Open standards
- Enable inter-organisational development

Continued on page 39...